18. Monday, July 29. **Connectedness.**

Last Time: · X is connected if $\forall\, U, V$ open with $X = U \cup V$ and $U \cap V = \emptyset$, either $U = \emptyset$ or $V = \emptyset$.

· $Y \subseteq X$ is connected if $\forall\, U, V \subseteq X$ open with $Y \subseteq U \cup V$ and $Y \cap U \cap V = \emptyset$, either $Y \subseteq U$ or $Y \subseteq V$.

Theorem 1: Let $f: X \to Y$ be a continuous function. If X is connected, then $f(X)$ is connected.

Proof: Since the corestriction $g = f|^{f(X)}: X \to f(X)$ is continuous, it suffices to consider the case of a continuous surjective map.

So assume $g: X \to Z$ is surjective continuous with X connected.

Let us show $Z = g(X)$ is also connected. We prove the contrapositive.

Assume $U, V \subseteq Z$ are open sets with $Z = U \cup V$ and $U \cap V = \emptyset$ with $U \neq \emptyset$ and $V \neq \emptyset$.

Since g is continuous, $g^{-1}(U)$ and $g^{-1}(V)$ are open in X.

Now, since $Z = U \cup V$, we have $X = g^{-1}(Z) = g^{-1}(U) \cup g^{-1}(V)$. We also have

$$g^{-1}(U) \cap g^{-1}(V) = g^{-1}(U \cap V) = g^{-1}(\emptyset) = \emptyset.$$

Finally, since g is surjective $U \neq \emptyset \Rightarrow g^{-1}(U) \neq \emptyset$ and $V \neq \emptyset \Rightarrow g^{-1}(V) \neq \emptyset$

Therefore $X = g^{-1}(U) \cup g^{-1}(V)$ is a separation of X $\square$

Corollary 2: If $\sim$ is an equivalence relation on a connected space X, Then The quotient space $X/_\sim$ is connected.

Proof: We have a continuous surjection $\pi: X \longrightarrow X/_\sim$ □

Theorem 3: Let X, Y TOP spaces. Then $X \times Y$ is connected $\Leftrightarrow$ X and Y are connected.

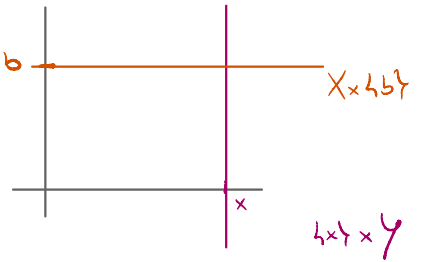
Proof: ($\Rightarrow$) If $X \times Y$ is connected, since we have continuous surjections

$$p_x : \underset{(x,y)}{X \times Y} \longrightarrow \underset{x}{X} \qquad \text{and} \qquad p_y : \underset{(x,y)}{X \times Y} \longrightarrow \underset{y}{Y}$$

It follows from Thm 1 That X and Y are connected.

($\Leftarrow$) Assume X, Y connected. We will use prop 17.3 (Union of connected $\cap \neq \emptyset$ is connected)
Fix your favorite $b \in Y$. For an arbitrary $x \in X$, consider The subspace

$$T_x = X \times \{b\} \cup \{x\} \times Y$$



Note That

1) $X \times \{b\}$ is connected, because $X \approx X \times \{b\}$
2) $\{x\} \times Y$ is connected, because $Y \approx \{x\} \times Y$
3) $X \times \{b\} \cap \{x\} \times Y = \{(x,b)\} \neq \emptyset$

Therefore, T_x is connected $\forall x \in X$.

Moreover, note that

$$X \times Y = \bigcup_{x \in X} T_x \quad \text{and} \quad \bigcap_{x \in X} T_x = X \times \{y\} \neq \emptyset.$$

Hence, again by Prop 17.3, $X \times Y$ is connected. □

Remark (HW): The same result holds for arbitrary products if one uses the product topology. However, $\mathbb{R}^{\mathbb{N}}$ is not connected with the box topology.

Connectedness in $\mathbb{R}$

Recall

1) Intermediate Value Theorem: Let $f: [a,b] \to \mathbb{R}$ be a continuous function. If $f(a) \leq f(b)$, then for every c with $f(a) \leq r \leq f(b)$, there exists some $c \in [a,b]$ such that $f(c) = r$.

2) Definition: Given a set $A \subset \mathbb{R}$, we say that $M \in \mathbb{R}$ is an upper bound for A if $a \leq M \quad \forall a \in A$.

3) Supremum Axiom: If $A \neq \emptyset$ admits an upper bound, then it admits a smallest upper bound, called the supremum of A, denoted $\sup A$